

STATE BOARD OF TECHNICAL EDUCATION AND TRAINING
TELANGANA

DIPLOMA EXAMINATION (C-21)

C21-END-NOV-24

SEMESTER III , SEMESTER END EXAM

AI/BM/CCB/CPS/CS/EC/EE/EI/ES/EV

EC-302

Digital Electronics



**PCODE
13023**

Exam Date: 12-11-2024

Session: FN

Duration: 2 Hours [11:30 AM To 01:30 PM]

[Total Marks: 40]

PART-A

Instructions:

1. Answer the following questions.
2. Each question carries **ONE** mark.

8 X 1 = 8

1. Define Fan - in capacity of a digital IC.
2. Define combinational logic circuit.
3. Define asynchronous counter.
4. Write the binary equivalent of hexadecimal number F2A0.
5. What is modulus of a counter?
6. State the need for a Register.
7. What is volatile memory?
8. What is the need of memory?

PART-B

Instructions:

1. Answer the following questions.
2. Each question carries **THREE** marks.

4 X 3 = 12

- 9(a). Draw basic gates with truth tables

---- OR ----

- 9(b). Draw the 4-bit bi-directional shift register circuit.

- 10(a). Draw the logic circuit and truth table of 1-to-4 demultiplexer.

---- OR ----

- 10(b). Draw the block diagram of A/D conversion using successive approximation method.

- 11(a). Draw the circuit of parallel-in parallel-out shift register.

11(b). A MOD-16 ripple counter is holding the count 1011. What will the count be after 16 clock pulses?

12(a). Define the terms access time, memory capacity, and word length with reference to memories

----- OR -----

12(b). State the need for flash memory.

PART-C

Instructions: 1. Answer the following questions.

4 X 5 = 20

2. Each question carries **FIVE** marks.

13(a). State and prove DeMorgan's theorems.

----- OR -----

13(b). Explain the working of a decade counter with circuit and timing diagram.

14(a). Explain the working of 4-bit parallel adder circuit using full adders.

----- OR -----

14(b). Draw the circuit diagram of instrumentation amplifier and mention its need(3+2)?

15(a). Explain the working of a 4-bit ring counter with timing diagram.

----- OR -----

15(b). Explain the working of 4-bit shift left register with a circuit and timing diagram.

16(a). Draw the circuit diagram of Binary weighted resistors type DAC and mention its disadvantages?(3+2)

----- OR -----

16(b). Draw and explain the circuit of A/D converter using counter method.